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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS – JULY / AUGUST 2025

Program	:	**B.E. – Information Science and Engineering**	**Semester**	:	**IV**
Course Name	:	**Microcontrollers**	**Max. Marks**	:	**100**
Course Code	:	**21IS44/ 22 / 23IS42**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	List and explain the four major design rules behind RISC processors implementation?	CO1	(08)
	b)	With a neat diagram explain an MCU chip, explain the internal components of a typical Microcontroller unit.	CO1	(06)
	c)	Explain how the ARM instruction set differs from the pure RISC instruction set?	CO1	(06)
2.	a)	Explain the internal components of system on chip. Write diagram.	CO1	(08)
	b)	Compare the RISC and CISC type of systems.	CO1	(06)
	c)	List the two modes of data transfer between MCU and peripherals? Explain the general functioning of interrupts.	CO1	(06)

UNIT - II

3.	a)	Why data alignment is important? What are the consequences of misaligned data access?	CO2	(05)
	b)	With a diagram, explain Inter integrated circuit (I2C) on-board protocol.	CO2	(08)
	c)	Write the classification of semiconductor memory types. Explain memory read and write cycle of SRAM chip.	CO2	(07)
4.	a)	Briefly explain the purpose of the following condition flags: i. Zero (Z) ii. Negative (N) iii. Carry (C) iv. Overflow (V)	CO2	(08)
	b)	Explain Load Store architecture used in RISC processors.	CO2	(06)
	c)	Discuss the operations of the descending stack in ARM7.	CO2	(06)

UNIT - III

5.	a)	Write a block diagram of a ARM7 Processor. List the modes of operation in ARM7.	CO3	(08)
	b)	Find the results of the following operations given the contents of R1=0x0089EF32, R2=0x80456730, R3=0x8 i. MOV R4, R2, LSR#4 ii. MVN R5, R1, ASR R3 iii. MVNS R6,R1	CO3	(06)

	c)	Discuss with an appropriate diagram for the register set working towards relationship between various modes of an ARM 32 bit and THUMB 16 bit processor.	CO3	(06)
6.	a)	Given R0=0x12345678, R1=0XD5678007 Find the contents of the destination registers, and the state of the relevant flags after execution of following instructions i. MVN R7, R0 ii. MOVN R5, R1 iii. ADDS R6, R0, R1	CO3	(08)
	b)	Write an ALP to add the first n even numbers. Store the result in a memory location.	CO3	(06)
	c)	Discuss on how the AMBA processor specifies the use of AHB & APB bus components.	CO3	(06)

UNIT- IV

7.	a)	Write an Assembly Language program to generate and print 10 Fibonacci numbers, excluding the first one(i.e. 0)	CO4	(08)
	b)	What is the content of R6 after execution of following instructions? Given that R6=0x44001100 and R2=0x04 i. LDR R2, [R6, #0056]! ii. STR R1, [R6, R2]! iii. LDR R3, [R6,LSL #4] Justify the results.	CO4	(06)
	c)	Write a C program to display the string 'MICROCONTROLLERS' in the serial window of UART1.	CO4	(06)
8.	a)	List the steps involved in execution of the following Instructions i. STMDB R1!{R7,R2-R4} if R1=0X44000078 ii. LDMIA R7!{R0-R5}	CO4	(08)
	b)	List the important features of LPC 214x series.	CO4	(06)
	c)	List and explain Registers used in timer/counter units in the polled mode.	CO4	(06)

UNIT - V

9.	a)	Explain the memory model of Cortex-MO and explain types and attributes of memory.	CO5	(10)
	b)	Explain Modes and states of Cortex- M0 processor.	CO5	(10)
10.	a)	Discuss the advantages and architectural features of Cortex-MO+.	CO5	(10)
	b)	List and Explain the interrupts of Cortex-MO and different states that each interrupts be in any time.	CO5	(10)

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